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Decision Table

Introduction

Decision points are the conditions or checkpoints in a software application where the system evaluates one or more inputs and decides the next action to perform. They play a vital role in Decision Table Testing by ensuring that all possible combinations of conditions and outcomes are considered during test design.

Key Points

- A decision point is a logical condition that determines the flow of execution in an application.
- It helps identify the actions that should be performed based on different input conditions.
- Decision points are represented using conditions such as **Yes/No, True/False, or Valid/Invalid.**
- They help testers create complete and systematic test cases by covering all possible business rules.
- Decision points reduce the chances of missing important test scenarios.
- They improve the accuracy, reliability, and quality of software testing.
- Decision points are widely used in banking systems, e-commerce websites, login modules, ATM systems, leave management systems, and online shopping applications.
- Each decision point may lead to one or more actions depending on the combination of conditions.
- They simplify the testing of complex business logic by organizing conditions and expected actions in a structured format.
- Decision points ensure that both positive and negative scenarios are verified before software is released.

Advantages of Decision Points

- Improve test coverage.
- Reduce redundant test cases.
- Help identify missing business rules.
- Make decision-making logic easier to understand.
- Increase software reliability and consistency.
- Support efficient and systematic testing.

Examples of Decision Points

- Is the username valid?

- Is the password correct?
- Is the account active?
- Is the product available in stock?
- Is the payment successful?
- Has the order been shipped?
- Is the leave approved by the manager?
- Is sufficient balance available in the bank account?

1. Online Shopping – Free Shipping Eligibility

CONDITIONS	RULES			
	R1	R2	R3	R4
C1: Order Amount $\geq$ ₹500	Y	Y	N	N
C2: User is Prime Member	Y	N	Y	N
C3: Product from Partner Seller	Y	N	Y	N
ACTIONS				
A1: Free Shipping	✓	✓	✗	✗
A2: Shipping Charges Applicable	✗	✓	✓	✓
A3: Show Eligible Message	✓	✗	✓	✗
A4: Show Not Eligible Message	✗	✓	✗	✓
Note: Y = Yes, N = No, ✓ = Action is executed, ✗ = Action is not executed				

Decision Table Conditions

- C1: Order Amount $\geq$ ₹500
- C2: Prime Member
- C3: Partner Seller

TC ID	Test Scenario / Inputs	Expected Output	Rule Covered
TC1_01	Order Amount = ₹700, Prime = Yes, Partner Seller = Yes	Free shipping applied	R1
TC1_02	Order Amount = ₹700, Prime = Yes, Partner Seller = No	Shipping charges applied	R2
TC1_03	Order Amount = ₹700, Prime = No, Partner Seller = Yes	Shipping charges applied	R3

TC1_04	Order Amount = ₹400, Prime = Yes, Partner Seller = Yes	Shipping charges applied	R4
TC1_05	Order Amount = ₹500, Prime = Yes, Partner Seller = Yes	Free shipping applied	R5
TC1_06	Order Amount = ₹300, Prime = No, Partner Seller = No	Shipping charges applied	R6

Individual Explanation of Each Test Case

TC1_01: Validates the happy path where all required conditions are satisfied. Free shipping should be applied successfully.

TC1_02: Verifies that products sold by non-partner sellers are not eligible for free shipping.

TC1_03: Ensures non-Prime users are charged shipping even when the order value exceeds ₹500.

TC1_04: Confirms that orders below ₹500 are not eligible for free shipping.

TC1_05: Tests the boundary value (₹500) to verify that free shipping starts exactly at the minimum eligible amount.

TC1_06: Verifies that when none of the conditions are satisfied, shipping charges are always applied.

2. User Login Validation

CONDITIONS	RULES			
	R1	R2	R3	R4
C1: Username Entered	Y	Y	Y	N
C2: Password Entered	Y	Y	N	Y
C3: Username Valid	Y	N	Y	N
C4: Password Valid	Y	N	Y	N
ACTIONS				
A1: Login Successful	✓	✗	✗	✗
A2: Show "Invalid Username"	✗	✓	✗	✗
A3: Show "Invalid Password"	✗	✗	✓	✗
A4: Show "Enter All Fields"	✗	✗	✗	✓
Note: Y = Yes, N = No, ✓ = Action is executed, ✗ = Action is not executed				

Decision Table Conditions

- C1: Username Valid
- C2: Password Valid
- C3: Account Active

Test Cases Derived from the Table

TC ID	Test Scenario / Inputs	Expected Output	Rule Covered
TC2_01	Username valid, Password valid, Account Active	Login successful	R1
TC2_02	Username valid, Password valid, Account Inactive	Account inactive message	R2
TC2_03	Username valid, Password invalid	Invalid password	R3
TC2_04	Username valid, Password invalid, Account Inactive	Invalid password displayed	R4
TC2_05	Username invalid	Invalid username	R5
TC2_06	Username invalid, Password invalid	Invalid username	R6

Individual Explanation of Each Test Case

TC2_01: Verifies successful login when all credentials are correct.

TC2_02: Ensures inactive accounts cannot access the application.

TC2_03: Confirms invalid passwords are detected correctly.

TC2_04: Verifies password validation occurs before checking account status.

TC2_05: Ensures invalid usernames are rejected immediately.

TC2_06: Confirms username validation has higher priority when multiple inputs are incorrect.

3. Bank ATM Cash Withdrawal

CONDITIONS	RULES				
	R1	R2	R3	R4	R5
C1: Card Inserted	Y	Y	Y	Y	N
C2: Correct PIN Entered	Y	Y	N	N	Y
C3: Sufficient Balance	Y	N	Y	N	-
C4: Daily Limit Exceeded	N	N	N	Y	-
ACTIONS					
A1: Cash Dispensed	✓	✗	✗	✗	✗
A2: Show "Invalid PIN"	✗	✓	✗	✗	✗
A3: Show "Insufficient Balance"	✗	✗	✓	✗	✗
A4: Show "Daily Limit Exceeded"	✗	✗	✗	✓	✗
A5: Show "Insert Card"	✗	✗	✗	✗	✓

Note:
Y = Yes, N = No, - = Not applicable,
✓ = Action is executed,
✗ = Action is not executed

Decision Table Conditions

- C1: Card Inserted
- C2: PIN Correct
- C3: Sufficient Balance
- C4: Daily Limit Exceeded

Test Cases Derived from the Table

TC ID	Test Scenario / Inputs	Expected Output	Rule Covered
TC3_01	Card inserted, Correct PIN, Balance available, Daily limit not exceeded	Cash dispensed	R1
TC3_02	Card inserted, Incorrect PIN	Invalid PIN message	R2
TC3_03	Card inserted, Correct PIN, Insufficient balance	Insufficient balance	R3

TC3_04	Card inserted, Correct PIN, Daily limit exceeded	Daily limit exceeded message	R4
TC3_05	No card inserted	Insert card message	R5
TC3_06	Card inserted, Incorrect PIN, Insufficient balance	Invalid PIN displayed	R6

Individual Explanation of Each Test Case

TC3_01: Validates successful withdrawal when all conditions are satisfied.

TC3_02: Ensures incorrect PIN prevents cash withdrawal.

TC3_03: Verifies the ATM checks account balance before dispensing cash.

TC3_04: Confirms withdrawal is blocked after reaching the daily limit.

TC3_05: Ensures the ATM does not proceed without card insertion.

TC3_06: Confirms PIN validation occurs before balance verification.

4. Leave Approval System

CONDITIONS	RULES					
	R1	R2	R3	R4	R5	R6
C1: Leave Days ≤ 2	Y	Y	N	N	-	-
C2: Leave Days $> 2 \ \& \ \leq 5$	N	N	Y	Y	-	-
C3: Leave Days > 5	N	N	N	N	Y	Y
C4: Manager Approved	Y	N	Y	N	Y	N
C5: HR Approved	-	-	Y	N	Y	N
ACTIONS						
A1: Leave Auto Approved	✓	✗	✗	✗	✗	✗
A2: Manager Approval Required	✗	✓	✓	✗	✗	✗
A3: HR Approval Required	✗	✗	✗	✓	✓	✓
A4: Leave Approved	✓	✗	✓	✗	✓	✗
A5: Leave Rejected	✗	✓	✗	✓	✗	✓
Note: Y = Yes, N = No, - = Not applicable, ✓ = Action is executed, ✗ = Action is not executed						

Decision Table Conditions

- C1: Leave ≤ 2 Days
- C2: Manager Approval
- C3: HR Approval

Test Cases Derived from the Table

TC ID	Test Scenario / Inputs	Expected Output	Rule Covered
TC4_01	Leave = 2 days	Leave approved automatically	R1
TC4_02	Leave = 4 days, Manager approved	Leave approved	R2
TC4_03	Leave = 4 days, Manager rejected	Leave pending/rejected	R3
TC4_04	Leave = 7 days, Manager approved, HR approved	Leave approved	R4
TC4_05	Leave = 7 days, Manager approved, HR rejected	Leave rejected	R5
TC4_06	Leave = 7 days, Manager rejected	Leave rejected	R6

Individual Explanation of Each Test Case

TC4_01: Verifies short leaves are approved automatically.

TC4_02: Ensures medium-duration leave requires manager approval.

TC4_03: Confirms leave is not approved without manager authorization.

TC4_04: Verifies long leave requires approvals from both manager and HR.

TC4_05: Ensures HR rejection overrides manager approval.

TC4_06: Confirms long leave is rejected if the manager does not approve.

5. E- Commerce Order Cancellation policy

CONDITIONS	RULES				
	R1	R2	R3	R4	R5
C1: Order Placed	Y	Y	Y	Y	Y
C2: Order Shipped	N	Y	Y	Y	N
C3: Delivery Completed	N	N	Y	Y	Y
C4: Cancellation Within 24 Hrs	Y	N	-	-	-
C5: Return Policy Applicable	-	-	Y	Y	N
ACTIONS					
A1: Cancel Order	✓	✗	✗	✗	✗
A2: Cancel Not Allowed	✗	✓	✓	✗	✓
A3: Initiate Return / Refund	✗	✗	✓	✓	✗
A4: Show "Order Already Delivered"	✗	✗	✗	✓	✓
A5: Show "Return Not Applicable"	✗	✗	✗	✗	✓

Note:
Y = Yes, N = No, - = Not applicable,
✓ = Action is executed,
✗ = Action is not executed

Decision Table Conditions

- C1: Order Placed
- C2: Order Shipped
- C3: Delivery Completed
- C4: Cancellation Within 24 Hours
- C5: Return Policy Applicable

Test Cases Derived from the Table

TC ID	Test Scenario / Inputs	Expected Output	Rule Covered
TC5_01	Order placed, Within 24 hours	Order cancelled successfully	R1
TC5_02	Order shipped	Cancellation not allowed	R2
TC5_03	Delivered, Return policy available	Return initiated	R3
TC5_04	Delivered, Return policy expired	Return rejected	R4
TC5_05	Order placed, Cancellation after 24 hours	Cancellation denied	R5
TC5_06	Order shipped and delivered	Cancellation not permitted	R6

Individual Explanation of Each Test Case

TC5_01: Validates successful cancellation before shipment within the allowed time period.

TC5_02: Confirms shipped orders cannot be cancelled.

TC5_03: Ensures customers can initiate returns when the return policy is valid.

TC5_04: Verifies returns are rejected after the return window expires.

TC5_05: Confirms cancellation requests after 24 hours are denied according to business rules.

TC5_06: Verifies that completed deliveries cannot be cancelled and must follow the return process instead.